

mike, real-world case studies...

Real-World Case Studies in Blockchain-Based Voting Systems

The transition of blockchain-based voting from theory to practice has been marked by diverse pilot programs. While these pilots aim to address challenges of transparency, auditability, and remote accessibility, they have also served as critical testing grounds for identifying security vulnerabilities. The following case studies analyze significant real-world applications of blockchain in municipal and federal electoral contexts.

Case Study 1: The Voatz Mobile Application (West Virginia, USA, 2018)

In the 2018 midterm elections, West Virginia became the first U.S. state to permit select voters—primarily military members and citizens living overseas—to cast federal ballots via the mobile application Voatz. The application utilized a permissioned blockchain (Hyperledger Fabric) to store an immutable record of the votes cast (Belfer Center, 2018).

Technical Implementation and Objectives

The primary objective was to improve voter turnout among disenfranchised overseas populations by providing a secure, remote alternative to traditional paper-based absentee ballots. The system combined biometric authentication (facial recognition) with a mobile-based blockchain ledger to ensure that votes remained tamper-proof once submitted.

Academic Critique and Security Findings

A security analysis by researchers at the MIT Computer Science and Artificial Intelligence Laboratory (CSAIL) identified significant vulnerabilities in the Voatz architecture. Specter et al. (2020) discovered that despite the use of blockchain for the final ledger, the mobile application itself was susceptible to attacks. Key findings included:

- **Vote Alteration:** An adversary with control over the user's device could alter or stop a user's secret ballot before it reached the blockchain (Specter et al., 2020).
- **Privacy Concerns:** Reliance on third-party services for facial recognition posed potential privacy risks (MIT News, 2020).
- **Transparency Issues:** The proprietary nature of the Voatz source code prevented independent public auditing (Specter et al., 2020).

Case Study 2: Municipal Consultative Pilot in Zug (Switzerland, 2018)

The Swiss city of Zug conducted a pioneering pilot for municipal voting between June and July 2018. This was a consultative vote designed to test the feasibility of integrating blockchain with a digital identity (eID) system (City of Zug, 2018).

Feature	Description
Identity Framework	U-Port digital identity based on the Ethereum blockchain.
Encryption	Paillier homomorphic encryption and zero-knowledge proofs (ZKP).
Participation	72 digital ID holders participated.
Citizen Approval	Over 75% of participants welcomed the introduction of e-voting.

Findings and Civic Impact

The pilot was deemed a success by municipal authorities regarding user experience and digital ID integration. The official evaluation report noted that 21% of surveyed voters believed blockchain improved security, while only 2% opposed the introduction of e-voting entirely (City of Zug, 2018).

Case Study 3: The Moscow City Duma Election Pilot (Russia, 2019)

In September 2019, Moscow implemented a blockchain-based internet voting system for three electoral districts. The source code was made available for public testing prior to the election (Gaudry, 2019).

Cryptographic Vulnerabilities

During testing, French researcher Pierrick Gaudry discovered that the system's 1024-bit ElGamal encryption keys were too small. Gaudry demonstrated that the private keys could be broken in approximately 20 minutes using standard hardware (Gaudry, 2019)..

BENEFITS AND CHALLENGES

BENEFITS AND CHALLENGES

Blockchain technology promises to transform the way governments operate and citizens vote. By distributing data across a tamper-resistant ledger, it offers a vision of transparent, auditable, and trustworthy democratic processes. However, this vision carries serious technical, social, and security trade-offs that must be evaluated honestly before large-scale adoption.

BENEFITS OF BLOCKCHAIN VOTING

1. Immutability & Tamper-Proof Records

Once a vote is recorded on a blockchain, it becomes cryptographically sealed within a chain of blocks. Altering any entry would require recalculating every subsequent block and controlling a majority of the network computationally infeasible in a well-designed system. This near-permanent integrity directly addresses traditional concerns around ballot stuffing, late-night result alterations, and vote destruction.

2. Remote Accessibility & Voter Inclusion

Blockchain-based voting systems can allow citizens to vote from any internet-enabled device eliminating geographic and physical barriers. This is particularly impactful for overseas military personnel, citizens with disabilities, rural populations, and diaspora communities who historically face disenfranchisement due to polling logistics.

3. End-to-End Transparency & Auditability

Public blockchains create a verifiable audit trail accessible to any authorized observer. Independent watchdogs, political parties, and even individual voters can verify that their vote was counted correctly without revealing who voted for whom through cryptographic proofs such as zero-knowledge proofs. This shifts election oversight from trust-based to mathematics-based.

4. Reduced Administrative Costs Over Time

Traditional elections require enormous expenditure: printing ballots, staffing polling stations, transporting ballot boxes, and manual counting. A mature blockchain voting infrastructure could substantially reduce these recurring costs, especially in nations with large geographic areas or dispersed populations.

CHALLENGES & CRITICAL RISKS

1. Severe Security Vulnerabilities: The Malware Problem

The blockchain ledger itself may be tamper-proof, but the devices voters use to cast their votes are not. Malware installed on a voter's phone or computer can silently alter or redirect a vote before it ever reaches the chain, a threat sometimes called the "last-mile problem." No cryptographic integrity of the ledger can protect against a compromised endpoint. Nation-state

actors, ransomware groups, and spyware vendors represent credible threats to any remote digital voting system.

2. The Digital Divide: Structural Inequity

Globally, roughly 2.7 billion people remain without reliable internet access. Within developed nations, elderly citizens, low-income communities, and rural populations are disproportionately offline. A blockchain voting system risks systematically silencing these groups, converting what was once a logistical inconvenience into a structural exclusion baked into the system's design.

3. Scalability Limits

Major public blockchains process a limited number of transactions per second far below what a national election requires. A U.S. federal election, for example, involves over 150 million votes within a compressed time window. Scaling blockchain networks to handle this throughput without sacrificing decentralization or security remains an unsolved engineering challenge, and private/permissioned blockchains introduced to fix this trade away the very trustlessness that makes the technology appealing.

4. Anonymity vs. Verifiability Tension

Democratic voting rests on two conflicting demands: votes must be secret (to prevent coercion), yet verifiable (so voters know their vote was counted). Cryptographic approaches like homomorphic encryption and zero-knowledge proofs offer partial solutions, but these are complex to implement correctly and even harder to audit or explain to the general public. A subtle flaw in implementation could compromise either anonymity or integrity.

Quick-Reference: Pros vs. Cons

Dimension	Benefit	Challenge
Data Integrity	Immutable, cryptographically sealed records	Endpoint/device malware bypasses ledger security
Access	Remote voting increases participation	Digital divide excludes offline populations
Transparency	Full audit trail verifiable by observers	Anonymity vs. verifiability tension
Cost	Long-term operational savings	High upfront development & infrastructure cost
Scale	Efficient for small-scale pilots	Throughput far below national election demands

IDEAS WORTH PURSUING

Rather than viewing blockchain as an all-or-nothing replacement for elections, the most promising path lies in hybrid and layered approaches. Here are ideas that balance innovation with realism:

Idea 1 Hybrid Blockchain + Paper Trail System

Every blockchain vote should generate a corresponding paper receipt that the voter can verify and deposit in a physical ballot box. This dual-record system preserves the auditability of digital voting while maintaining a human-readable fallback that courts and election officials can examine. Estonia, which has run internet voting since 2005, uses a risk-layered model that could be further strengthened with this approach.

Idea 2 Permissioned Consortium Chain for Election Authorities

Rather than a fully public blockchain, a consortium of trusted institutions election commissions from multiple states or countries, international observers, and civil society organizations — could co-govern a permissioned blockchain. Each participant holds validator authority, and no single government controls the ledger. This preserves decentralization at the institutional level while achieving the throughput and accountability that public chains cannot.

Idea 3 Zero-Knowledge Proofs for Voter Verification

Zero-knowledge proofs (ZKPs) allow a voter to cryptographically prove they are eligible and that their vote was counted without revealing their identity or their choice. Integrating ZKPs into the vote-casting protocol would be a significant leap toward solving the anonymity-verifiability paradox. Advances in zk-SNARKs and zk-STARKs are making this increasingly practical, and funding research in this area would pay long-term democratic dividends.

Idea 4 Blockchain for Voter Registration Only (Staged Adoption)

A lower-risk entry point is using blockchain not for vote casting but for voter roll management. Voter registration data on a blockchain would be resistant to purges, duplication, and politically motivated manipulation. Citizens could verify their own registration status in real time, and election officials from different jurisdictions could share a synchronized, auditable record. This achieves real democratic gains with far less security exposure than full online voting.

Idea 5 Community-Controlled Nodes in Underserved Areas

To address the digital divide, blockchain voting infrastructure could include community validation nodes housed in libraries, community centers, and schools particularly in rural and low-income areas. Citizens without home internet access could vote on secured terminals that connect to the chain locally, bridging the gap between digital innovation and equitable access. Node operators from the community would also serve as grassroots election observers.

Closing Perspective

Blockchain is not a silver bullet for democratic integrity but it is a genuinely powerful tool when applied with precision. The greatest risk is not the technology itself; it is the temptation to deploy it prematurely at scale, driven by novelty rather than readiness. The path forward is incremental, transparent, and deeply attentive to who gets left behind. Democracy's strength has always been its legitimacy in the eyes of all citizens and any technology that claims to protect democracy must first prove it works for everyone.

maraki

Application of Blockchain Technology in Voting Systems

The application of blockchain technology to electoral systems provides a decentralized, mathematically verifiable architecture designed to resolve the systemic vulnerabilities inherent in traditional voting methods. While historical voting systems rely on centralized authorities to distribute, collect, and tally ballots—creating single points of failure susceptible to tampering, human error, and voter suppression—blockchain introduces a trustless environment. In this context, "trustless" does not mean a lack of trust, but rather that the system's integrity relies on cryptographic proofs and distributed consensus rather than a central governing body. The translation of traditional voting mechanics into a blockchain framework relies heavily on three core technological pillars: cryptographic keys for identity verification, smart contracts for election execution, and public distributed ledgers for immutable record-keeping.

Cryptographic Keys and Identity Verification

A primary challenge in digital voting is the paradox of verifiable anonymity: the system must simultaneously prove that a voter is eligible and authorized to vote, while ensuring their specific ballot choice remains entirely secret. Blockchain addresses this through asymmetric cryptography, utilizing pairs of public and private cryptographic keys.

In a blockchain-based voting system, the election commission acts as a registration authority prior to the election. Upon verifying a citizen's real-world identity (via national ID, biometrics, or other KYC protocols), the system issues a digital voting token or authorization right to the voter's secure digital wallet. The voter's public key acts as their pseudonymous address on the blockchain, while their private key acts as their digital signature. When a voter casts their ballot, they sign the transaction using their private key. The network can mathematically verify that the signature corresponds to a registered public key without exposing the private key. To further separate the voter's real-world identity from their ballot, advanced cryptographic techniques such as Zero-Knowledge Proofs (ZKPs) can be implemented. ZKPs allow the system to verify that an individual possesses the right to vote and has cast a valid ballot without revealing the voter's identity or the contents of the vote to the public ledger.

Smart Contracts and Election Automation

The operational logic of the election is managed by smart contracts—self-executing code deployed on the blockchain that automatically enforces the rules of an agreement. In a voting application, a smart contract acts as an incorruptible digital ballot box and automated tallying mechanism.

The smart contract is programmed with the specific parameters of the election: the start and end times, the list of eligible public keys, and the candidate options. Because the smart contract resides on the blockchain, its code is transparent and auditable by any third party prior to the election, ensuring no malicious logic (such as vote-flipping) is present. Once the election begins,

the smart contract receives incoming vote transactions. It automatically checks each transaction against its programmed rules: it verifies the cryptographic signature, checks if the public key is on the whitelist of eligible voters, and ensures the timestamp falls within the active election window. If these conditions are met, the smart contract accepts the vote and updates the tally in real-time. Because the tallying is automated through consensus-verified code, the need for manual, centralized ballot counting is eliminated, drastically reducing the potential for human error or internal fraud.

The Public Ledger and Immutability

Once a vote is processed by the smart contract, it is packaged with other transactions into a "block." Through a consensus mechanism (such as Proof of Stake or Proof of Authority), the network of decentralized nodes validates the block and appends it to the distributed ledger. This public ledger serves as a transparent, chronological, and immutable database of all cast votes.

The immutability of the blockchain is secured through cryptographic hashing. Each block contains a unique hash of its own data and the hash of the preceding block. If a malicious actor attempts to alter a single vote in a past block, it would change the hash of that block, subsequently breaking the cryptographic link to all following blocks. The decentralized network would immediately reject this altered version of the ledger. Consequently, once a vote is successfully recorded on the blockchain, it cannot be deleted, altered, or replaced. This provides a permanent, universally verifiable audit trail, allowing individual voters to verify that their specific ballot was included in the final tally without revealing their choice to others.

Solving the Double-Voting Problem

A critical concern in any digital voting system is the digital equivalent of ballot stuffing: double-voting. In cryptocurrency, this is known as the "double-spend" problem, where a user attempts to spend the same digital token twice. Blockchain's underlying architecture naturally solves this issue for voting.

When a voter casts their ballot, the transaction utilizes the specific voting token or right allocated to their wallet. As the transaction propagates through the peer-to-peer network, nodes verify the ledger's history. If the ledger shows that the voter's public key has already executed a successful vote transaction, any subsequent attempts to cast a vote using the same credentials will be automatically rejected by the network's consensus protocol and the smart contract logic. This real-time, network-wide validation ensures that the fundamental democratic principle of "one person, one vote" is strictly and securely enforced at the cryptographic level.

APPROVAL CERTIFICATE

E-Flow Document Management System

Document: mjjj
Generated: 5/23/2026, 4:11:12 PM
Document ID: 46

APPROVAL CHAIN

1

staff

Approved: 5/23/2026, 4:11:12 PM

HMAC: cfdb25ebba84c8848a9308ab3fdd06a7006521e...

Drawn Signature:

